Eduardo Sampaio Pimenta

Mathematician — Probability Theory and Stochastic Processes

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Research interests

Probability theory, stochastic processes, scaling limits, invariance principles, boundary random walks, Feller processes, dynamic percolation and evolving random environments.

Research group

PROFE — Grupo de Pesquisa em Probabilidade e Física Estatística

Member of the research line *Interacting Particle Systems*, registered in the CNPq Directory of Research Groups.

Academic experience and education

- **Postdoctoral fellow**, Federal University of Minas Gerais (UFMG),Belo Horizonte, Brazil, 2026–present.
  Host: Remy de Paiva Sanchis.
- **Postdoctoral fellow**, University of Augsburg, Augsburg, Germany, 2025–2026.
  Host: Markus Heydenreich.
- **PhD in Mathematics**, Federal University of Bahia (UFBA), Salvador, Brazil, 2021–2025.
  Advisors: Tertuliano Franco and Dirk Erhard.
- **MSc in Mathematics**, Federal University of Bahia (UFBA), Salvador, Brazil, 2019–2021.
  Advisor: Tertuliano Franco.
- **BSc in Mathematics**, Federal University of Minas Gerais (UFMG), Belo Horizonte, Brazil, 2010–2017.

Publications and preprints

1. D. Erhard, T. Franco, M. Jara and E. Pimenta. "A Functional Central Limit Theorem for the General Brownian Motion on the Half-Line." Accepted, forthcoming in *Bernoulli*. Journal page.

2. J. C. Arroyave, E. Barros and E. Pimenta. "Scaling Limit of Boundary Random Walks: A Martingale Problem Approach." Preprint, 2025. arXiv:2507.10528.

Theses

- **Functional Central Limit Theorem and Berry–Esseen Estimates for Non-Homogeneous Random Walks**. Doctoral thesis, UFBA, 2025. Advisor 1: Tertuliano Franco; Advisor 2: Dirk Erhard.
- **Construction of the Dawson–Watanabe Process**. Master's dissertation, UFBA, 2021. Advisor: Tertuliano Franco.

Selected talks and presentations

- "Scaling Limit of Boundary Random Walks: A Martingale Problem Approach." XVII CLAPEM, Montevideo 2026.
- "A Functional Central Limit Theorem and Weak Berry–Esseen Estimates for Non-Homogeneous Random Walks." IHES Summer School — Statistical Aspects of Nonlinear Physics, Paris 2025.
- "Scaling Limits of Boundary Random Walks to Feller-Type Processes on the Half-Line." Augsburg 2025.
- "A Functional CLT for the General Brownian Motion on the Half-Line." IX Encontro da pos-graduacao em matematica da UFBA, Salvador 2024.
- "Berry–Esseen Estimates Based on Generators with Applications." VIII Encontro da Pós-graduação da UFBA, Salvador 2023.

Last updated: November 6, 2026. This is a preliminary version to be completed from the full academic CV.